Limits to improving reproduction number estimates by increasing case counts: Environmental variability, reporting heterogeneity, and surveillance design

Chi Wang

[a]*School of Safety Science and Engineering, Xi'an University of Science and Technology, Xi'an, 710054, China*

Abstract

Estimation errors in the reproduction number are frequently attributed to sample size. Here we show that the return on increasing case counts depends fundamentally on the inferential target, distinguishing the conditional instantaneous reproduction number R_t from the cross-period mean $\bar{R}$. Under negative-binomial transmission and binomial reporting, the estimation error of R_t decays as $1/m$ with expected reporting scale m, attaining the Cramér–Rao bound. For single-period estimators of $\bar{R}$ that are conditionally unbiased for R_t, however, the estimation error retains an irreducible variance floor v_R arising from environmental stochasticity, with reporting heterogeneity introducing an additional component v_H into our estimator's variance when reporting probabilities fluctuate across periods. This irreducibility is specific to expanding single-period case volume and does not preclude variance reduction through accumulating independent observation periods. We derive the variance balance scale $m_\times = A/v_R$, which characterises the point of diminishing sample returns without prescribing a minimum operational threshold, and reduce surveillance optimisation to three levers: sample volume, period count, and reporting standardisation. Analysing 10,303 valid state-weeks of influenza hospitalisations from the US National Healthcare Safety Network (2020–2026), we find that early-epidemic squared error scales affinely with inverse sample size (intercept $a = 0.0991$, 95% CI 0.0837–0.1165; slope $b = 2.243$, 1.494–3.007; ratio $b/a = 22.6$, 13.5–34.9). Theoretical analysis indicates that expanding single-period sample volume yields diminishing returns; empirical observations are compatible with this error structure, though current data cannot separate a strict non-zero asymptotic plateau from slow decay or uniquely attribute the fitted intercept to environmental

variance. Reporting adjustments account for $< 1\%$ of the intercept excess. Surveillance design should distinguish outbreak detection from estimation of mean transmission, and balance additional case capture against temporal replication, spatial pooling, and reporting standardisation according to the inferential target and dependence structure.

Keywords: Reproduction number, Environmental stochasticity, Reporting heterogeneity, Surveillance design, Variance decomposition, Influenza hospitalisations

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1. Introduction

The time-varying reproduction number R_t is a cornerstone metric for tracking epidemic dynamics and guiding public health interventions (Wallinga and Teunis, 2004; Fraser, 2007; Cori et al., 2013). An estimate exceeding unity indicates an expanding outbreak, whereas values below unity signal contraction. While extensive methodological research has refined the statistical estimation of R_t from given incidence time series—addressing sliding window choices, reporting delays, generation interval distributions, and truncation adjustments (Thompson et al., 2019; Gostic et al., 2020; Nouvellet et al., 2018; Steyn et al., 2025)—surveillance system designers face an antecedent question: how many infections must be captured, across how many facilities and periods, to attain a designated level of inferential precision?

This question is acute in severe-outcome surveillance, such as hospitalisation monitoring. Severe cases represent a small and variable fraction of total infections—typically 0.6%–2% for seasonal influenza (Zhou et al., 2012; Tokars et al., 2018). In systems such as the US National Healthcare Safety Network (NHSN), weekly hospital admissions in smaller jurisdictions frequently number only dozens of cases, and reporting protocols may alternate between mandatory and voluntary regimes (Centers for Disease Control and Prevention, 2026). As multi-model collaborative ensembles and scoring rules are deployed for real-time forecasting (Reich et al., 2019; Cramer et al., 2022; Mathis et al., 2024; Bosse et al., 2023), surveillance authorities require quantitative guidance on resource allocation: what fraction of estimation error can be eliminated by expanding patient volume, whether raising reporting completeness is preferable to broadening population coverage, and at what scale returns on sampling begin to diminish.

Classical renewal theory offers an deceptively straightforward answer. Under a Poisson renewal process, the Fisher information for the reproduction number grows linearly with observed case counts, implying that estimation variance scales inversely with sample volume (Parag et al., 2022). Superspreading and overdispersion in secondary infections, typically parameterised through a negative-binomial offspring distribution (Lloyd-Smith et al., 2005; Endo et al., 2020; Kucharski et al., 2020), increase the variance prefactor but preserve its $1/m$ scaling rate. Under this framework, expanding case volume yields smooth, monotonic precision gains, with no structural scale beyond which single-period sampling becomes unremunerative.

In real-world surveillance, two distinct noise mechanisms invalidate this

intuition. The first is environmental stochasticity. Hosts in the same jurisdiction and time interval share exogenous transmission conditions, including meteorology, school calendars, and population mobility (Dalziel et al., 2018; Rasmussen et al., 2011). Ecology distinguishes environmental stochasticity, which affects all individuals simultaneously, from demographic stochasticity, which operates independently across individuals (Engen et al., 1998; Rohani et al., 2002). While the relative contribution of demographic stochasticity vanishes as sample size increases, environmental stochasticity does not attenuate with population volume. Increasing case counts refines our estimate of the instantaneous transmission potential R_t in a specific environment, but cannot eliminate the cross-period variation of R_t itself. When surveillance aims to infer the underlying typical transmission potential across periods or to forecast the subsequent generation, environmental variance forms an irreducible error component under single-period scale expansion.

This issue is conceptually distinct from the difference between true and apparent reproduction numbers elucidated by Eales and Riley (2024). Eales and Riley (2024) demonstrated that an apparent reproduction number reconstructed from downstream surveillance proxies (such as hospitalisations or deaths) is delayed and smoothed relative to the true transmission potential due to convolution with delay kernels, even when case counts are infinite and observation is deterministic. In our framework, the two inferential targets are the conditional instantaneous value and the cross-period mean; while both are affected by convolution when using lagged surveillance counts to construct empirical growth statistics (Section 6.3), environmental variance is an intrinsic attribute of biological transmission rather than an artefact of deconvolution.

The second noise mechanism arises from reporting heterogeneity. The fraction of cases recorded by surveillance fluctuates across periods due to changing healthcare-seeking behaviour, diagnostic testing availability, and institutional compliance. Such fluctuations act multiplicatively across all cases in a given period and cannot be averaged out by sampling more patients within that interval. Prior contributions have quantified the information content of epidemic curves and developed state-space models and filtering techniques for time-varying transmission (Azmon et al., 2014; Stoner et al., 2019; Bracher and Held, 2021; Bhatt et al., 2023; Flaxman et al., 2020; Parag, 2021; Parag and Donnelly, 2022; Parag et al., 2025). However, existing formulations have not integrated environmental and reporting variances into an explicit design framework, leaving unanswered where sampling returns dimin-

ish, how reporting noise interacts with environmental fluctuations, and how surveillance budgets should be balanced across sample volume, observation periods, and data harmonisation.

Here, we formulate a negative-binomial transmission and binomial observation framework that formally separates two inferential targets: the conditional instantaneous reproduction number R_t (Target 1) and the marginal cross-period mean $\bar{R}$ (Target 2). For Target 1, the natural estimator $\widehat{R}_t = C_t/m$ is conditionally unbiased and attains the Cramér–Rao lower bound, with variance decaying strictly as $1/m$. For Target 2, any conditionally unbiased estimator retains an environmental variance floor v_R, yielding a total variance $V(m) = v_R + A/m$. We derive the variance balance scale $m_\times = A/v_R$, which identifies the scale where sampling and environmental variances balance. We extend this decomposition to account for reporting heterogeneity v_H, demonstrate an inherent unidentifiability between environmental and reporting variances from count moments alone, and characterise an analytical reporting signature that permits mechanistic diagnosis. Finally, we establish optimal budget allocation rules across sample volume, temporal replication, and reporting standardisation, and assess whether the predicted scaling patterns are compatible with US influenza hospitalisation data and cross-pathogen surveillance datasets.

2. Background and theoretical framework

2.1. Transmission, reporting, and the two estimation targets

Consider transmission during discrete generation period t. Following Lloyd-Smith et al. (2005), we model superspreading and individual-level transmission heterogeneity by assuming that, conditional on the instantaneous reproduction number R_t, each of the n primary infected individuals generates a random number of secondary infections $X_{i,t}$ drawn independently from a negative-binomial offspring distribution:

$$X_{i,t} \mid R_t \sim \mathrm{NB}\left(k,\ \frac{k}{k+R_t}\right), \qquad \mathbb{E}(X_{i,t} \mid R_t) = R_t, \quad \mathrm{Var}(X_{i,t} \mid R_t) = R_t + \frac{R_t^2}{k}, \tag{1}$$

where $k > 0$ is the overdispersion parameter; smaller values of k denote heavier overdispersion, and $k \to \infty$ recovers Poisson transmission. The total incidence of secondary infections in the generation is $I_t = \sum_{i=1}^{n} X_{i,t}$. By

infinite divisibility and the convolution property of the negative-binomial family, conditional on R_t and n, total infections follow

$$I_t \mid n, R_t \sim \mathrm{NB}\left(nk,\ \frac{k}{k+R_t}\right), \qquad \mathbb{E}(I_t \mid n, R_t) = nR_t, \quad \mathrm{Var}(I_t \mid n, R_t) = nR_t + \frac{nR_t^2}{k}. \tag{2}$$

Surveillance typically captures only a fraction of total infections. We model case observation through binomial thinning: conditional on total infections I_t, each infection is independently confirmed and reported with probability $\rho \in (0, 1]$, yielding the observed case count $C_t \sim \mathrm{Bin}(I_t, \rho)$. A well-known preservation property of the negative-binomial distribution under binomial thinning ensures that the reported count retains an exact negative-binomial form:

$$C_t \mid n, R_t \sim \mathrm{NB}\left(nk,\ \frac{k}{k+\rho R_t}\right), \qquad \mathbb{E}(C_t \mid n, R_t) = \rho n R_t, \quad \mathrm{Var}(C_t \mid n, R_t) = \rho nR_t + \frac{\rho^2 nR_t^2}{k}. \tag{3}$$

We define $m = \rho n$ as the expected reporting volume, which serves as the natural scale parameter for the surveillance system.

We formulate the analysis under the following foundational assumptions:

H1 (Binomial observation): Conditional on I_t, each infection is ascertained independently with a time-invariant reporting probability $\rho \in (0, 1]$.

H2 (Homogeneous branching): Conditional on R_t, primary infections generate secondary infections independently and identically according to Eq. (1).

H3 (Exogenous overdispersion): The dispersion parameter k is an invariant characteristic of the pathogen and host population, independent of R_t and n.

H4 (Predetermined ancestor scale): The active ancestor volume n is fixed by study design (e.g., in active contact tracing) and is statistically independent of R_t.

The actual transmission potential R_t fluctuates over time. Meteorology, school vacations, public holidays, and behavioral shifts drive continuous variation around a baseline level (Dalziel et al., 2018; Rasmussen et al., 2011).

We denote the marginal expectation and environmental variance by

$$\mathbb{E}(R_t) = \bar{R}, \qquad \mathrm{Var}(R_t) = v_R. \tag{4}$$

In ecological population biology, fluctuations shared by all co-circulating lineages within a time interval are termed environmental stochasticity, distinguishing them from individual-level demographic stochasticity (Engen et al., 1998; Rohani et al., 2002).

This distinction defines two fundamentally different estimation targets:

- **Target 1 (Conditional instantaneous reproduction number, R_t):** The realised transmission potential in the current period given the ambient environment. This target is relevant for real-time situational awareness, outbreak detection, and evaluating immediate intervention efficacy.
- **Target 2 (Marginal cross-period mean, $\bar{R}$):** The central tendency of transmission potential across periods. Under temporal independence of environmental conditions, $\bar{R}$ also constitutes the minimum mean squared error one-step-ahead forecast for the next generation.

Both targets share the natural plug-in estimator $\widehat{R}_t = C_t/m = C_t/(\rho n)$. However, their estimation errors differ fundamentally in their reference benchmark: the error for Target 1 is $\widehat{R}_t - R_t$, whereas the error for Target 2 is $\widehat{R}_t - \bar{R} = (\widehat{R}_t - R_t) + (R_t - \bar{R})$, which carries an additional, sample-invariant environmental fluctuation term $R_t - \bar{R}$.

Proposition 1 (Estimation of Target 1)**.** *Under H1–H4, for any given ancestor scale n and reporting probability ρ, the estimator $\widehat{R}_t = C_t/m$ satisfies:*

(i) It is conditionally unbiased for R_t: $\mathbb{E}(\widehat{R}_t \mid R_t) = R_t$.

(ii) Its conditional variance decays strictly as $1/m$:

$$\mathrm{Var}(\widehat{R}_t \mid R_t) = \frac{R_t}{\rho n} + \frac{R_t^2}{kn} = \frac{R_t}{m} + \frac{\rho R_t^2}{km}. \tag{5}$$

(iii) It attains the Cramér–Rao lower bound: the conditional Fisher information for R_t is $\mathcal{I}(R_t) = [R_t/m + \rho R_t^2/(km)]^{-1}$, and $\widehat{R}_t$ is the unique uniformly minimum-variance unbiased estimator (UMVUE).

Proposition 1 confirms that for Target 1, increasing sample volume m continuously and monotonically suppresses estimation variance toward zero, consistent with classical information-theoretic scaling (Parag et al., 2022).

2.2. Variance decomposition and the balance scale

For Target 2, the behavior of $\widehat{R}_t$ departs sharply from the $1/m$ scaling law.

Proposition 2 (Estimation variance and balance scale for Target 2)**.** *Under H1–H4, the marginal variance of $\widehat{R}_t$ around $\bar{R}$ decomposes into:*

$$V(m) = \mathrm{Var}(\widehat{R}_t \mid n) = v_R + \frac{\bar{R}}{\rho n} + \frac{\bar{R}^2 + v_R}{kn} = v_R + \frac{A}{m}, \tag{6}$$

where the scale-dependent coefficient A is

$$A = \bar{R} + \frac{\rho(\bar{R}^2 + v_R)}{k}. \tag{7}$$

Defining the variance balance scale $m_\times = A/v_R$, the error variance satisfies:

(i) ***Asymptotic floor:*** *As $m \to \infty$, $V(m) \to v_R > 0$.*

(ii) ***Variance partition:*** *The sampling share of total variance is $1/(1 + m/m_\times)$; at $m = m_\times$, the sampling variance exactly equals the environmental variance floor.*

(iii) ***Diminishing returns to scale doubling:*** *Doubling sample volume from m to $2m$ reduces total variance by a fraction of $1/[2(1 + m/m_\times)]$.*

(iv) ***Elasticity:*** *$V(m)$ is strictly decreasing and strictly convex, with scale elasticity $\epsilon_m = -m_\times/(m + m_\times)$, which equals $-1/2$ at $m = m_\times$.*

(v) ***Optimality bound:*** *In the class of estimators conditionally unbiased for R_t, $V(m)$ is the minimum mean squared error for $\bar{R}$. Under H4 with R_{t+1} independent of past history $\mathcal{F}_t$, it is also the minimum mean squared prediction error for one-step-ahead growth $y = C_{t+1}/m$.*

We emphasize that the balance scale $m_\times = A/v_R$ serves solely as a comparative benchmark where sampling and environmental variances are of equal magnitude; it does not prescribe a minimum operational threshold satisfying specific estimation precision or outbreak detection power requirements.

Equation (6) follows directly from the law of total variance: the expected conditional variance yields $A/m = \mathbb{E}[1/\mathcal{I}(R_t)]$, whereas the variance of the

conditional expectation yields v_R. For any estimator $\tilde{\theta}$ conditionally unbiased for R_t, the cross-term in $\mathbb{E}[(\tilde{\theta} - R_t)(R_t - \bar{R})]$ vanishes identically:

$$\mathbb{E}(\tilde{\theta} - \bar{R})^2 = \mathbb{E}\big[\mathrm{Var}(\tilde{\theta} \mid R_t)\big] + v_R \geq v_R + \mathbb{E}\big[1/\mathcal{I}(R_t)\big] = v_R + \frac{A}{m}, \quad (8)$$

with equality attained by $\widehat{R}_t = C_t/m$. For the one-step prediction result, we additionally assume that R_{t+1} is independent of $\mathcal{F}_t$. Under this condition, any predictor $\hat{y}$ based on history $\mathcal{F}_t$ forecasting $y = C_{t+1}/m$ satisfies

$$\mathbb{E}(\hat{y} - y)^2 \geq \mathrm{Var}(y \mid \mathcal{F}_t) = v_R + \frac{A}{m}, \quad (9)$$

with equality attained by $\hat{y} = \bar{R}$.

Because $V(m)$ is strictly convex with no inflection points, $m_\times$ provides a natural dimensional reference (Fig. 1B): when $m \ll m_\times$, error is dominated by demographic and reporting sampling noise; when $m \gg m_\times$, error is dominated by environmental stochasticity, and further sample expansion faces rapidly diminishing returns. For typical baseline parameters ($\bar{R} = 1.15$, $v_R = 0.04$, $k = 0.35$, $\rho = 0.25$), we have $A = 2.1232$ and $m_\times = 53.1$ cases. At sample volumes of $m = 10$, $m_\times = 53.1$, and $m = 200$, doubling sample size reduces total variance by 42.1%, 25.0%, and 10.5%, respectively (Fig. 1C).

Remark 1 (Reporting completeness versus population coverage)**.** *Expected reporting volume $m = \rho n$ can be increased either by raising reporting completeness ρ or by expanding population coverage n. Holding n fixed and driving $\rho \to 1$ yields an asymptotic variance limit of $v_R + \bar{R}/n + (\bar{R}^2 + v_R)/(kn)$, which remains strictly greater than v_R. Conversely, holding ρ fixed and expanding $n \to \infty$ drives variance to the theoretical floor v_R. Increasing n by a small percentage yields a variance reduction that is $A/\bar{R} = 1 + \rho(\bar{R}^2 + v_R)/(k\bar{R})$ times larger than achieving the same percentage gain via ρ.*

The lower bound v_R applies strictly to single-period, conditionally unbiased estimators under temporally independent environments ($\phi = 0$). When R_t follows a stationary AR(1) process with autocorrelation ϕ, historical information can be borrowed, reducing the irreducible one-step-ahead environmental component to $(1 - \phi^2)v_R$ (yielding $0.75v_R$ for $\phi = 0.5$ and $0.36v_R$ for $\phi = 0.8$). Similarly, pooling J jurisdictions with identical environmental variance v_R and average pairwise cross-jurisdiction correlation $\bar{c}$ reduces the environmental variance to

$$v_R\,\bar{c} + \frac{v_R(1 - \bar{c})}{J}. \quad (10)$$

When local disturbances are imperfectly correlated ($\bar{c} < 1$), spatial pooling systematically compresses the environmental floor; when shocks are perfectly synchronised ($\bar{c} = 1$), spatial aggregation provides no reduction.

2.3. Reporting heterogeneity and identifiability

When reporting probability fluctuates across time, an additional noise component enters the error floor. We replace H1 with:

H5 (Stochastic reporting): The reporting probability $p_t \in (0, 1]$ is a stationary stochastic process independent of R_t and n, with $\mathbb{E}(p_t) = \rho$, $\mathrm{Var}(p_t) = \sigma^2$, relative volatility $\tau = \sigma/\rho$, and standardised rate $u_t = p_t/\rho$.

H6 (Conditional independence): Given trajectories $(R_1, \ldots, R_T)$ and $(p_1, \ldots, p_T)$, observed counts across periods are mutually independent.

Proposition 3 (Variance decomposition under reporting heterogeneity)**.** *Under H2–H6, denoting $U = (1 + \tau^2)(\bar{R}^2 + v_R)$:*

(i) The marginal variance of $\widehat{R}_t = C_t/m$ decomposes into three components:

$$V_H(m) = \mathrm{Var}(\widehat{R}_t \mid n) = v_R + v_H + \frac{A_H}{m}, \tag{11}$$

where the reporting heterogeneity variance v_H and modified scale coefficient A_H are

$$v_H = \tau^2(\bar{R}^2 + v_R), \qquad A_H = \bar{R} + \frac{\rho U}{k}. \tag{12}$$

(ii) The generalised balance scale becomes $m_{\times,H} = A_H/(v_R + v_H) < m_\times$.

(iii) As $m \to \infty$, $V_H(m) \to v_R + v_H$.

(iv) ***Unidentifiability:*** *From the marginal distribution of $(C_1, \ldots, C_T)$ up to second moments, (v_R, v_H) are unidentifiable; only the compound parameter $U = (1 + \tau^2)(\bar{R}^2 + v_R)$ is identified.*

(v) ***Reporting signature:*** *When evaluating growth ratios $G_t = C_t/D_t$ with denominator $D_t = \sum_s w_s C_{t-s}$ under stationary AR(1) reporting*

shocks with persistence ψ, the prediction error intercept depends on lag weights via

$$F(w,\psi) = 1 - 2\sum_{s} w_s \psi^s + \sum_{s}\sum_{s'} w_s w_{s'} \psi^{|s-s'|}, \tag{13}$$

generating diagnostic intercept ratios that differentiate reporting noise from transmission dynamics.

Under baseline parameters with $\tau = 0.15$, reporting heterogeneity introduces $v_H = 0.0307$ and compresses the balance scale from $m_\times = 53.1$ down to $m_{\times,H} = 30.4$ (Fig. 1D).

Corollary 1 (Ranking reversal between reporting completeness and coverage)**.** *When deciding whether to allocate resources to expanding population coverage n or raising reporting completeness ρ:*

(i) If reporting fluctuations are institutional and scale-invariant ($\mathrm{d}\tau^2/\mathrm{d}\rho = 0$), expanding coverage n strictly dominates increasing ρ.

(ii) If reporting fluctuations stem from binomial sampling where $\mathrm{Var}(p_t) = \rho(1-\rho)/(\nu+1)$ with sample size ν, the ranking reverses: raising ρ strictly dominates expanding n when

$$-\rho\,\frac{\mathrm{d}\tau^2}{\mathrm{d}\rho}\,(kn+1) > 1+\tau^2 \quad \Longleftrightarrow \quad m > \frac{\rho^2\nu}{k}. \tag{14}$$

For $\nu = 100$ and $k = 0.35$, the reversal occurs at $m > 17.9$ cases; for $\nu = 1000$, it occurs at $m > 178.6$ cases (Fig. 1E).

3. Results

3.1. The scale balance point for a moving target

The variance curves derived in Section 2.2 quantify the balance point of sampling returns. Taking baseline transmission parameters ($\bar{R} = 1.15$, $v_R = 0.04$, $k = 0.35$, $\rho = 0.25$), the sampling coefficient is $A = 2.1232$, yielding a variance balance scale of $m_\times = 53.1$ cases. At sample volumes of $m = 10$, 53.1, and 200, doubling the surveillance scale reduces total variance by 42.1%, 25.0%, and 10.5%, respectively. While the error variance for Target 1 (R_t) decays strictly as $1/m$ without saturation, the error for Target 2 ($\bar{R}$) is

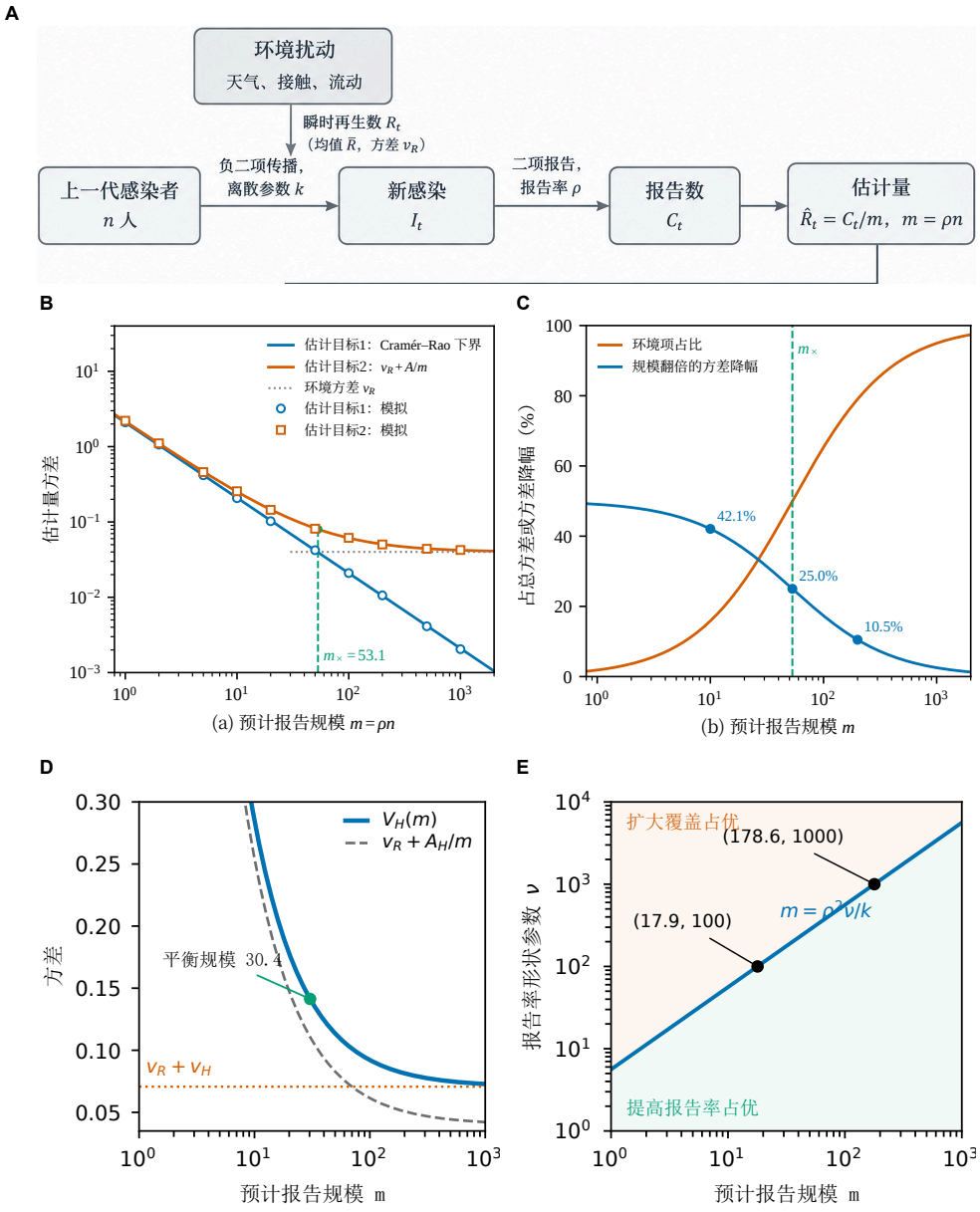


Figure 1: Model structure and theoretical variance decomposition. (A) Transmission and observation process: environmental disturbances determine the instantaneous reproduction number R_t; n primary cases generate secondary infections I_t via negative-binomial branching; infections undergo binomial thinning with probability ρ into reported counts C_t; bottom row contrasts Target 1 (R_t) and Target 2 ($\bar{R}$). (B) Variance of the natural estimator $\widehat{R}_t = C_t/m$ as a function of expected reporting scale m, evaluated at baseline parameters ($\bar{R} = 1.15, v_R = 0.04, k = 0.35, \rho = 0.25$; $A = 2.1232, m_\times = 53.1$ cases); hollow circles denote 20,000 Monte Carlo replications; green dashed line marks the variance balance scale $m_\times$. (C) Proportion of environmental variance in total error and proportional variance reduction from scale doubling as a function of m; dotted markers highlight reductions of 42.1%, 25.0%, and 10.5% at $m = 10, 53.1, 200$. (D) Variance curve under reporting heterogeneity ($\tau = 0.15$), where the balance scale contracts from 53.1 to 30.4 cases; dotted horizontal line denotes the asymptotic floor $v_R + v_H$. (E) Ranking reversal boundary between raising reporting completeness and expanding population coverage under binomial reporting volatility.

bounded from below by v_R; once m surpasses $m_\times$, additional sample volume compresses only the sampling term, which already accounts for less than half of total variance (Fig. 1B).

When surveillance capacity is constrained by total budget rather than single-period sample volume, planners must determine how to partition resources over time.

Proposition 4 (Multi-period allocation and cost optimization)**.** *Under H1–H4, consider observing T periods with expected reporting volume m per period, yielding a total volume $B = Tm$. We estimate $\bar{R}$ by the cross-period average $T^{-1}\sum_{t=1}^{T}\widehat{R}_t$. Let c_0 denote the fixed operational cost per period, c_1 the marginal cost per reported infection, and $\mathcal{B}$ the total budget, such that $T = \mathcal{B}/(c_0 + c_1 m)$.*

(i) If environmental conditions R_t are independent across periods, the estimation variance is

$$\frac{1}{T^2}\mathrm{Var}\Big(\sum_{t=1}^{T}\widehat{R}_t\Big) = \frac{v_R}{T} + \frac{A}{B}. \tag{15}$$

The sampling term A/B depends solely on total volume B, invariant to temporal partitioning, while the environmental term decays as $1/T$. If R_t follows a stationary Gaussian AR(1) process with autocorrelation ϕ, the sampling term remains A/B, while the environmental term satisfies

$$T^2\mathrm{Var}(\bar{R}_T) = v_R\Big[T + 2\sum_{h=1}^{T-1}(T-h)\phi^h\Big] = v_R\Big[\frac{T(1+\phi)}{1-\phi} - \frac{2\phi(1-\phi^T)}{(1-\phi)^2}\Big]. \tag{16}$$

(ii) In both settings, variance decreases strictly monotonically with T. Under temporal independence, the budget-constrained variance $f(m) = (c_0 + c_1 m)(v_R + A/m)/\mathcal{B}$ achieves its global minimum at

$$m^* = \sqrt{\frac{c_0}{c_1}}\, m_\times, \qquad f(m^*) = \frac{(\sqrt{c_0 v_R} + \sqrt{c_1 A})^2}{\mathcal{B}}. \tag{17}$$

If $\mathcal{B} < c_0 + c_1 m^*$, the boundary constraint $T \geq 1$ binds, yielding $T^* = 1$; if $c_0 = 0$, $m^* \to 0$ and $T^* \to \infty$.

The optimal single-period volume is the geometric mean of the cost ratio c_0/c_1 and the balance scale $m_\times$, isomorphic to Neyman–Cochran allocation in two-stage sampling (Cochran, 1977). In our empirical scenarios (Section 3.3), $m_\times$ spans 21.3–49.2 cases, implying $m^* \in [14.6, 22.2]$ for $c_0/c_1 = 10$ and $m^* \in [46.1, 70.1]$ for $c_0/c_1 = 100$.

For outbreak detection (Target 1), the inferential task is hypothesis testing: $H_0 \colon R_t \leq 1$ versus $H_1 \colon R_t > 1$. Because the negative-binomial family for C_t has a monotone likelihood ratio in R_t, the uniformly most powerful test rejects H_0 when $C_t \geq c_\alpha$, where c_α is the critical value at nominal size α (Karlin and Rubin, 1956; Lehmann and Romano, 2005). The exact statistical power at $R_t = 1 + \delta$ is

$$\beta(n, \delta) = 1 - F_{\mathrm{NB}}(c_\alpha - 1;\ nk,\ k/[k + \rho(1 + \delta)]). \tag{18}$$

Standard Wald tests comparing $(\widehat{R}_t - 1)/\sqrt{V_0}$ to normal quantiles exhibit substantial size inflation at small scales (e.g., actual size 7.81% for $m = 5, k = 0.1$; peak size 10.40% for $k = 1$). Due to discreteness, power curves exhibit non-monotonic serration with sample volume (e.g., dropping 82 times as n grows from 400 to 699 at $k = 0.35, \delta = 0.3$). Under $\rho = 0.25, k = 0.35$, reliably detecting a 15% transmission increase ($\delta = 0.15$) with 80% power requires $n \geq$ 2,107 active ancestors ($m \approx 527$ cases), approximately ten times the balance scale $m_\times$ (Supplementary Fig. S3). Thus, early-warning surveillance volume must be sized by statistical power rather than by $m_\times$.

Epidemic renewal simulations (Section 6.4) confirm that these scale relationships hold when incidence grows dynamically. Grouped by realized denominator scale D_t, simulated mean squared errors are within 0.98–1.03 times the single-generation reference curve $(v_R + A/m)/\bar{R}^2$ for $D_t \geq 400$, while random denominator noise slightly elevates error (1.04–1.23 times) at small scales $D_t < 400$ (Fig. 4E–F).

3.2. The reporting signature in the scaling law

Because environmental fluctuations and reporting noise both contribute to the scale-invariant intercept, distinguishing them requires examining how estimation errors respond to denominator lag weighting. We evaluate five competing structural mechanisms: M-env (stationary environmental stochasticity only); M-bin0 (binomial reporting without environmental variance); M-meas (pure observation measurement error); M-nodn (noise-free denominator); and M-mult (multiplicative reporting shocks on top of environmental noise).

Under stationary reporting shocks with autocorrelation ψ, the model-implied intercept ratio between lag-weight vectors w and w' follows $F(w,\psi)/F(w',\psi)$ from Eq. (13). For our primary weighting $w = (0.65, 0.25, 0.10)$, current-week weighting $w^{(1)} = (1, 0, 0)$, and three-week equal weighting $w^{(3)} = (1/3, 1/3, 1/3)$, pure reporting noise implies analytical intercept ratios of $q_1 \in [0.883, 1.338]$ and $q_2 \in [0.892, 1.374]$ (Supplementary Table S2).

Empirical estimation across these three weighting schemes yields slope estimates of $b = 2.243$, 3.134 (2.250–4.013), and 1.926 (1.083–2.858), with slope ratios relative to baseline of $r_1 = 1.397$ (1.229–1.623) and $r_2 = 0.859$ (0.599–1.083). The fitted intercept increases monotonically with average look-back lag (0.0682, 0.0991, and 0.1524 across mean lags of 1.0, 1.45, and 2.0 weeks), producing intercept ratios of $q_1 = 0.688$ (0.658–0.727) and $q_2 = 1.539$ (1.420–1.656). In 2,000 state-clustered bootstrap replications, 100% satisfied $q_1 < 1 < q_2$ (Supplementary Fig. S1).

While slope ratios rule out the noise-free denominator model (M-nodn), they do not separate reporting noise from environmental stochasticity. The observed intercept pattern $q_1 < 1 < q_2$ decisively rejects stationary environmental models ($q = 1$), but is consistent with epidemic growth dynamics: daily-resolution renewal simulations with admission delays reproduce $q_1 \in [0.601, 0.887]$ and $q_2 \in [1.187, 1.568]$ even under independent environments ($\phi = 0$). Thus, intercept ratios differentiate dynamic epidemic growth from stationary models, but cannot isolate reporting shocks from convolution delays without auxiliary reporting data (Fig. 2C–D).

3.3. A case study of influenza hospitalisations in the United States

We analyse weekly laboratory-confirmed influenza hospital admissions across 50 US states and Washington, D.C., from NHSN (2020–2026), comprising 16,218 state-weeks and 10,303 valid observations ($D_t \geq 5$). Because influenza's generation interval is approximately 3 days, weekly hospitalisation growth G_t aggregates roughly two infection generations. We therefore test the functional scaling shape rather than treating coefficients as structural single-generation parameters. The primary analysis focuses on the early expansion phase across three epidemic seasons (N = 1,267 state-weeks, mean growth $\bar{R}_{\text{early}} = 1.6706$, standard deviation 0.7116). Partitioned into six bins of expected scale D_t, mean squared error drops from 0.329 (95% CI 0.244–0.415) in the $[5, 20)$ bin to 0.161 in $[20, 50)$, 0.112 in $[100, 250)$, and levels off at 0.103 and 0.105 in the largest bins (Fig. 2A–B).

Feasible generalised least squares (FGLS) yields an estimated intercept of $a = 0.0991$ (95% CI 0.0837–0.1165), a slope of $b = 2.243$ (1.494–3.007), and an empirical ratio of $b/a = 22.6$ cases/week (13.5–34.9; BCa CIs 0.0819–0.1144, 1.538–3.090, and 14.2–36.6). Across 2,000 bootstrap replications, no sample produced $a \leq 0$ or $b \leq 0$. Ordinary least squares yielded a consistent slope of $b = 2.305$. Trimming the top 1% most influential loss observations reduced b/a to 10.6 (5.9–17.3), reflecting sensitivity to extreme outliers.

Empirical observations display diminishing error reductions with increasing sample volume, consistent with theoretical error decomposition. Model comparisons on mean functional shape (Supplementary Table S1) demonstrate that a floorless model $L = b/D$ performs substantially worse than the affine model (leave-one-state-out cross-validation deviance difference 1949.9± 243.1; 48 of 51 states favor the affine form). A floorless power-law model $L = cD^{-p}$ yields $\hat{p} = 0.273$, with a cross-validation deviance difference of 56.8 ± 27.2 relative to the affine specification. While the data require an error component that does not vanish with sample volume, current empirical ranges cannot differentiate a strictly flat asymptotic floor from slow power-law decay.

Decomposition of sample loss reveals the remaining scope for scale expansion. Average squared error across the sample is 0.181, of which the intercept a accounts for 54.6%, the scale-dependent term b/D_t accounts for 45.1% (evaluated at the harmonic mean scale of 27.4), and residual variation accounts for 0.2%. In a counterfactual scenario where all jurisdictions with $D_t < 250$ are expanded to $D_t = 250$, mean loss decreases by 40.9% to 0.107, closely matching the empirical average loss of 0.104 observed among state-weeks naturally exceeding 250 cases.

The empirical scale ratio aligns in magnitude with theoretical scenarios. Setting $\bar{R} = 1.67$, $k \in [0.2, 0.5]$ (Lloyd-Smith et al., 2005), $\rho \in [0.0056, 0.021]$ (Zhou et al., 2012; Tokars et al., 2018), and $v_R \in [0.04, 0.08]$, implied $m_\times$ spans 21.3–49.2 cases. Implied scenario intercepts span 0.0143–0.0287, and slopes span 0.610–0.707. The empirical intercept is 3.5–6.9 times larger, and the slope is 3.2–3.7 times larger, than scenario baselines; because both parameters are inflated by similar factors, their ratio falls within the theoretical range. The fitted intercept reflects unmodelled error components not explained by sample volume and cannot be simply equated with biological environmental variance.

3.4. Reporting adjustment and cross-pathogen replication

Proposition 3 establishes that count data alone cannot attribute intercept elevation to environmental variance versus reporting volatility. NHSN weekly hospital reporting completeness provides an observable empirical proxy. In early-epidemic weeks, reporting completeness volatility is modest (V_π = 0.00118, 95% CI 0.00092–0.00171; mean $\pi_{s,t}$ = 1.006), whereas voluntary reporting regimes exhibited volatility two orders of magnitude higher (V_π = 0.119).

Adjusting for hospital reporting completeness leaves the estimated intercept essentially unchanged: unadjusted $a = 0.0991$ (0.0820–0.1146) versus adjusted $a^* = 0.0994$ (0.0821–0.1150), yielding $\Delta a = a - a^* = -0.0004$ (-0.0014 to 0.0006). In two one-sided tests for equivalence (TOST) with a 5% margin (± 0.0050), the 90% BCa interval $[-0.0012, 0.0005]$ falls entirely within bounds. The upper confidence limit of 0.00065 represents only 0.93% of the scenario excess intercept ($a - 0.08/\bar{R}^2 = 0.0704$). Hence, observable institutional reporting completeness accounts for $< 1\%$ of the non-sampling error floor (Fig. 3A–B).

Cross-pathogen analysis across seven pathogen-window combinations in NHSN (influenza, COVID-19, RSV) demonstrates consistent directional behavior: all seven combinations exhibit $q_1 < 1 < q_2$ in 100% of bootstrap samples. Based on pre-specified classification criteria against renewal null ranges, intercept ratios do not reject dynamic growth nulls. However, intercept excess $a - 0.08/\bar{R}^2$ yields evidence of partial replication: the lower confidence bound exceeds zero for influenza calendar, growth, and out-of-sample seasons, and RSV 2024–2025 (4 partial replications), while COVID-19 windows and RSV growth windows show directional consistency (3 directional). Overall grading yields 0 full replications, 4 partial replications, and 3 directional findings (Fig. 3C–E). Independent outpatient ILINet surveillance (N = 1,479) and hospital FluSurv-NET ($N = 364$) corroborate the directional signature ($q_1 = 0.687, q_2 = 1.467$).

3.5. Design levers and budget allocation

Surveillance design must also account for the precision with which the environmental floor v_R can be estimated.

Proposition 5 (Estimation precision of environmental variance)**.** *Under H1–H4 with equal sample volume m per period and $Y_t = C_t/m$:*

(i) The modified moment estimator

$$\hat{v}_R = \frac{S_Y^2 - \varpi(\bar{Y})}{1 + \rho/(km)}, \qquad \varpi(y) = \frac{y}{m} + \frac{\rho y^2}{km}, \tag{19}$$

where $\bar{Y}, S_Y^2$ are sample mean and variance across T periods, has a variance that asymptotically plateaus as $m \to \infty$ at Ω_∞/T, where $\Omega_\infty = \mathbb{E}(R_t - \bar{R})^4 - v_R^2$. Expanding single-period sample volume cannot push estimation variance below this floor; the floor height is governed strictly by period count T.

(ii) The Fisher information lower bound on relative standard error is $(0.505T)^{-1/2}$, invariant to sample volume m ($0.81, 0.45, 0.20$ for $T = 3, 10, 50$).

(iii) Under stationary Gaussian AR(1) environments with autocorrelation ϕ, the effective period count for variance estimation contracts to

$$T_{\text{eff}} = \frac{T}{\kappa_T(\phi^2)} \approx T\,\frac{1 - \phi^2}{1 + \phi^2}. \tag{20}$$

Corollary 2 (Season-level components)**.** *If transmission potential includes an exchangeable seasonal shock $\eta_s \sim \mathcal{N}(0, v_S)$ across S seasons, the relative standard error of any unbiased estimator of v_S is bounded below by $\sqrt{2/S}$ ($\sqrt{2/(S-1)}$ when mean is estimated), regardless of weekly sample volume or jurisdiction count.*

For $S = 3$ and $S = 4$ seasons, the lower bounds are 0.82 and 0.71, respectively, demonstrating that resolving seasonal transmission variance requires multi-year observation horizons.

The division of labor among surveillance design levers is captured by the T-period error decomposition:

Proposition 6 (Roles of sample volume, period count, and standardization)**.** *Under H1–H5, with serial correlations $\gamma_R(h)$ and $\gamma_u(h)$:*

$$\mathbb{E}(\bar{R}_T - \bar{R})^2 = \frac{1}{T^2}\sum_{s=1}^{T}\sum_{t=1}^{T}\Big\{v_R\gamma_R(h) + \tau^2\gamma_u(h)\big[\bar{R}^2 + v_R\gamma_R(h)\big]\Big\} + \frac{A_H}{Tm}. \tag{21}$$

Increasing sample volume m reduces solely the final term; increasing T suppresses the environmental term and decaying reporting shocks; standardizing reporting ($\tau = 0$) eliminates the reporting term.

In baseline numerical evaluations ($T = 10, m = 50, \tau = 0.15, \phi = \psi = 0.5$), mean squared error is 22.5×10^{-3} (environmental 10.40, reporting 7.85, sampling 4.29). A tenfold expansion in sample volume lowers MSE only to 18.7×10^{-3}; quadrupling period count lowers it to 6.2×10^{-3}; eliminating reporting volatility reduces it to 14.6×10^{-3} (Fig. 4C).

When environments are positively autocorrelated ($\phi > 0$), consecutive periods provide redundant information:

Proposition 7 (Optimal surveillance design under AR(1) environments)**.** *Under budget constraint $T(c_0 + c_1 m) = \mathcal{B}$, there exists a threshold budget $\mathcal{B}^\dagger(\phi)$. When $\mathcal{B} < \mathcal{B}^\dagger$, the optimal strategy sets fewer, larger periods ($m^* > m_0^*$); when $\mathcal{B} > \mathcal{B}^\dagger$, optimal design sets more, smaller periods ($m^* < m_0^*$). As $\mathcal{B} \to \infty$, $m^* \to m_0^* \sqrt{(1-\phi)/(1+\phi)}$.*

For $\phi = 0.8$ and $c_0/c_1 = 100$, the budget threshold is $\mathcal{B}^\dagger = 775$. Sizing surveillance using naive independent-period formulas under small budgets results in severely undersized single-period sample volumes (Fig. 4D).

3.6. Seasonal and phase-dependent error

Empirical estimation confirms strong season-level heterogeneity. Individual season intercepts are 0.071 (0.049–0.094), 0.028 (0.017–0.041), and 0.135 (0.111–0.166), with between-season heterogeneity test $Q = 47.4$ ($p = 5 \times 10^{-11}$). Including the out-of-sample 2025–2026 season ($a = 0.177$) increases Q to 85.0 ($p = 2.6 \times 10^{-18}$). Between-season variance is 0.0041, with wide random-effects intervals (-0.056 to 0.211) reflecting the $\sqrt{2/(S-1)}$ precision bound for $S = 3$ (Fig. 5B).

Epidemic phase transitions introduce substantial benchmark error. During peak and post-peak weeks (N = 1,220), mean growth falls to $\bar{R}_{\text{peak}} = 0.8075$. Evaluating error against the early-phase baseline $\bar{R}_{\text{early}} = 1.6706$ yields large, scale-invariant losses (0.29–0.31 across bins), with benchmark bias $(\bar{R}_{\text{early}} - \bar{R}_{\text{peak}})^2 / \bar{R}^2_{\text{early}} = 0.2669$ explaining 88.9% of total loss. Rebenchmarking against peak-phase mean restores scale responsiveness: $a = 0.0665$ (0.0467–0.0897), $b = 7.927$ (5.424–11.225), and ratio $b/a = 119.2$ (Fig. 5C–D).

Shifts in reporting regimes highlight structural reporting shocks. Standard deviation of growth ratios increased by 192% (from 0.624 to 1.823) during the voluntary reporting transition, returning to 0.649 upon resumption of mandatory reporting. Stratification reveals that dispersion expansion

occurred specifically during the in-season transition month (October 2024; IQR difference 1.107, 0.289–1.430), while off-season dispersion remained stable (Fig. 5E–F).

Robustness analyses corroborate the positive scaling coefficient across specifications ($b/a \in [10.6, 124.7]$). Two-stage instrumental variable regression using lagged scale instruments yields $b_{\text{iv}} = 2.53$ (1.11–4.11; first-stage $F = 1147.1$), confirming that endogeneity does not drive the scale effect. National aggregate time series ($\bar{D} \approx 6{,}662$ cases/week) exhibit an average loss of 0.0348—one-third of state-level intercepts—consistent with spatial correlation $\bar{c} \approx 0.34$. External FluSight hospital ensemble forecasts ($a = 0.1295, b = 7.805, b/a = 60.3$) and European COVID-19 data ($b = 33.92, b/a = 124.7$) display consistent scaling directions (Supplementary Fig. S2).

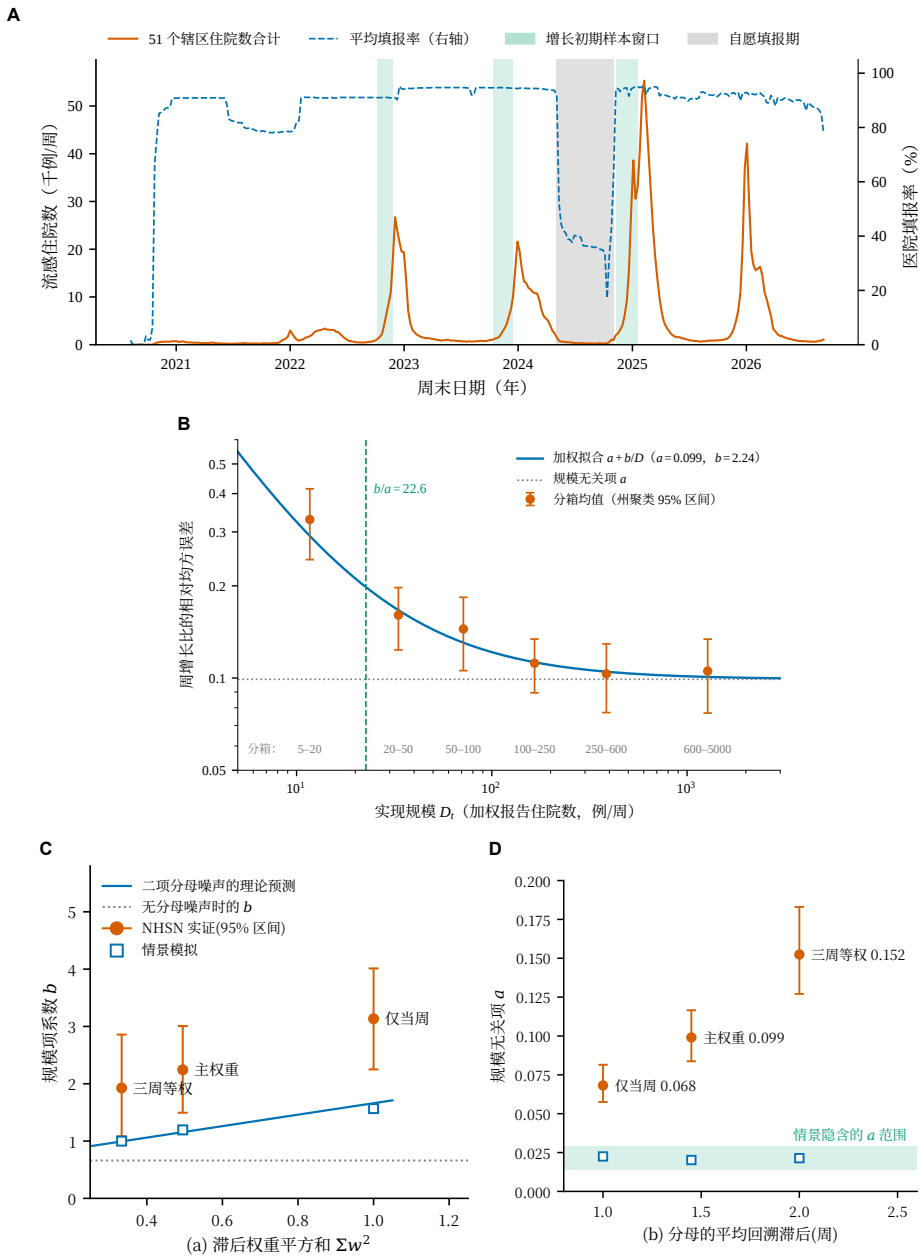


Figure 2: Scaling law and reporting signature in US influenza hospitalisation surveillance. (A) National weekly reported influenza hospitalisations and seasonal delineation (NHSN 2020–2026). (B) Binned mean squared loss of weekly hospitalisation growth against inverse sample scale $1/D_t$ in early epidemic phases, with FGLS affine fit $L = a + b/D_t$ (dashed curve); point area reflects state-week sample size. (C) Scaling slope coefficient b versus sum of squared lag weights $\sum_s w_s^2$ across three weighting schemes; points with error bars represent NHSN estimates and 95% CIs, squares denote scenario renewal simulations, solid line is the theoretical binomial denominator-noise prediction, and dotted line indicates the noise-free denominator baseline. (D) Estimated intercept a versus average denominator lookback lag, with scenario-implied intercept bounds shown in shaded grey.

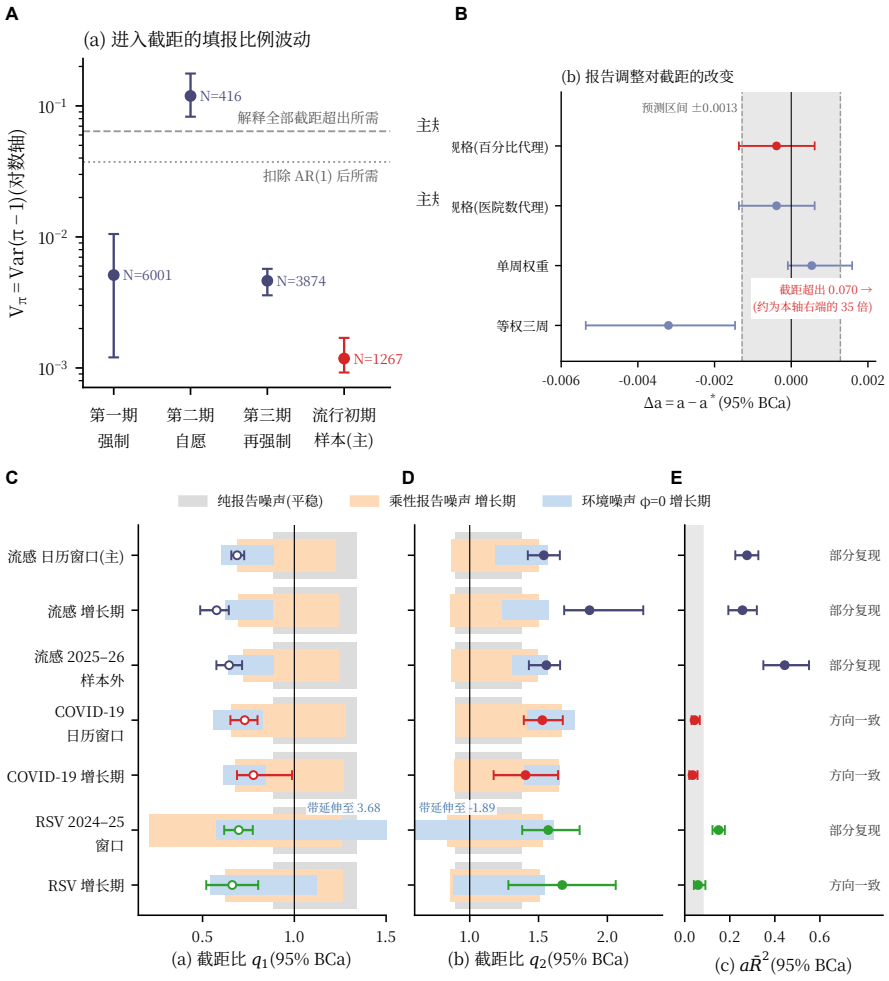


Figure 3: Reporting adjustment and cross-pathogen replication. (A) Volatility of hospital reporting completeness ratios V_π across three institutional reporting regimes and early-epidemic samples (logarithmic axis); horizontal reference lines indicate the reporting volatility required to explain the entire intercept excess or residual excess. (B) Difference in estimated intercept $\Delta a = a - a^*$ following hospital reporting adjustment across weighting vectors, with shaded band denoting pre-specified equivalence bounds (± 0.0050). (C) Empirical intercept ratios q_1 and (D) q_2 across seven pathogen-window combinations in NHSN; grey bands indicate analytical pure reporting noise bounds, and orange/blue bands represent simulated renewal null ranges. (E) Implied biological variance $a\bar{R}^2$ if intercepts were attributed entirely to environmental stochasticity, with right-hand labels denoting pre-specified replication classifications.

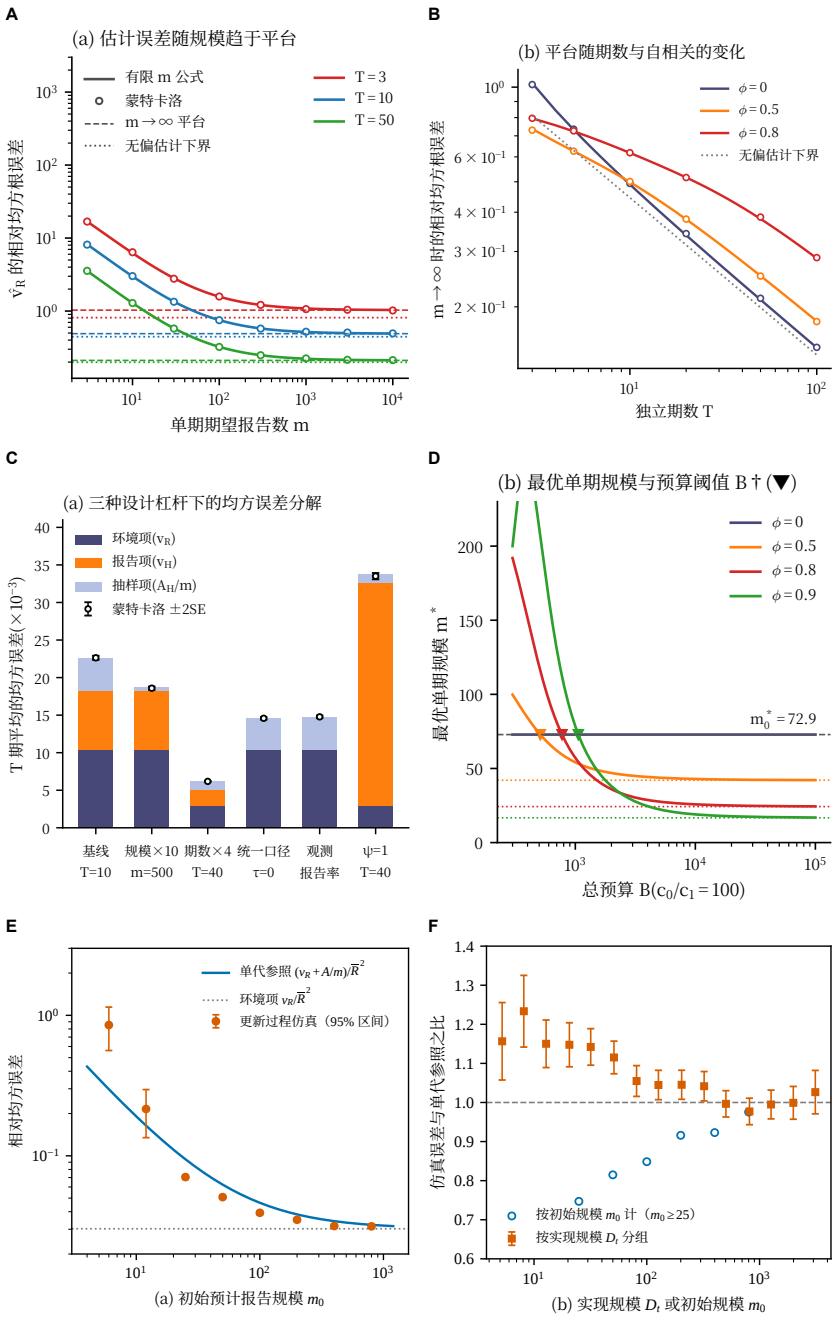


Figure 4: Estimation precision of environmental variance and surveillance design trade-offs. (A) Relative root mean squared error of the modified moment estimator $\hat{v}_R$ versus single-period sample volume m across $T = 3, 10, 50$ observation periods; solid curves show analytical finite-scale formulas, circles denote 20,000 simulations, dotted lines indicate asymptotic $m \to \infty$ plateaus, and dashed lines mark Fisher information lower bounds. (B) Asymptotic variance plateau height versus period count T and environmental autocorrelation ϕ. (C) Three-way variance decomposition of T-period average error under varying surveillance levers (scale expansion, temporal replication, reporting standardization), with circles showing 60,000 simulations (± 2 SE). (D) Budget-constrained optimal single-period scale m^* versus total budget $\mathcal{B}$ under AR(1) environments; triangles indicate the critical budget threshold $\mathcal{B}^\dagger$, dashed line marks the independent optimum $m_0^* = 72.9$, and dotted line is the asymptotic limit. (E) Relative mean squared error of growth statistics in epidemic renewal simulations versus initial scale m_0, compared against the single-generation

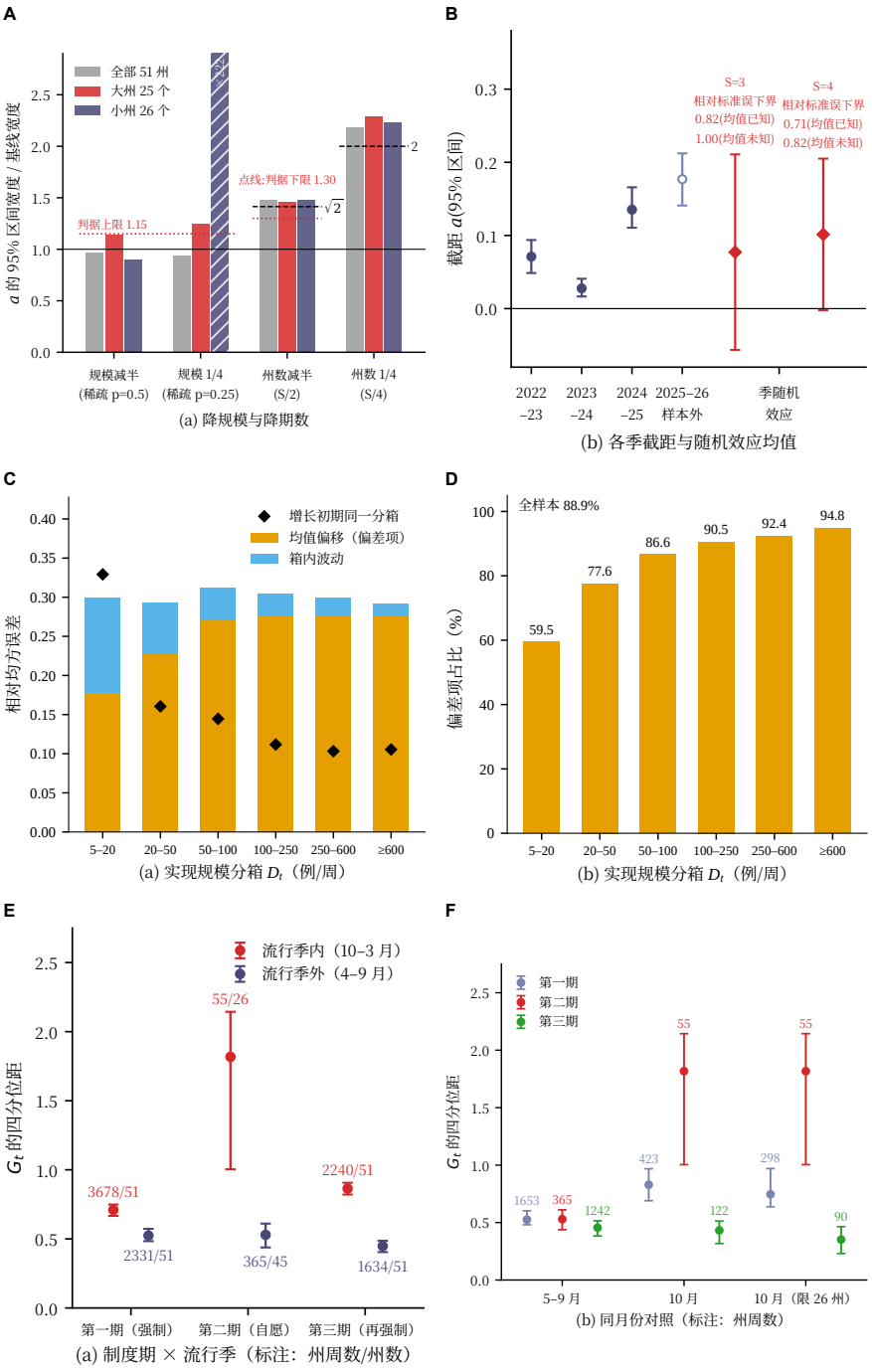


Figure 5: Season- and phase-dependent error structures in influenza surveillance. (A) Multiplier of 95% confidence interval width for intercept a under binomial thinning (sample size reduction) and state subsampling (period count reduction); dashed curve indicates the $\sqrt{S/S'}$ theoretical scaling expectation. (B) Season-specific intercepts and random-effects mean, with hollow circle denoting the out-of-sample 2025–2026 season and red labels indicating theoretical relative standard error bounds. (C) Decomposition of squared error across scale bins during peak and post-peak weeks, separating benchmark shift bias (orange) from within-bin variance (blue); diamonds denote early-phase loss. (D) Proportion of total loss explained by benchmark bias across scale bins. (E) Interquartile range of weekly growth ratios G_t across three institutional reporting regimes, stratified by in-season and off-season weeks. (F) Month-matched control of growth ratio dispersion, isolating the

4. Discussion

The return on expanding case volume in epidemic surveillance depends fundamentally on the inferential target. For the conditional instantaneous reproduction number R_t, returns to scale do not saturate: estimation variance decays strictly as $1/m$, attaining the Cramér–Rao lower bound (Proposition 1). For the cross-period mean $\bar{R}$, however, estimation error contains two components that do not attenuate with sample volume: environmental variance v_R, arising from shared meteorological, social, and mobility fluctuations across periods; and reporting heterogeneity variance v_H, arising from temporal instability in ascertainment (Propositions 2 and 3). The variance balance scale $m_\times = A/v_R$ (compressed to $A_H/(v_R + v_H)$ under reporting noise) defines the scale where sampling noise balances the environmental floor. Parag and colleagues (Parag et al., 2022; Parag and Donnelly, 2022) characterized the sampling-dependent component from an information-theoretic perspective; our results show that environmental stochasticity introduces an extrinsic floor that operates independently of population volume, providing a surveillance-design analogue to classical ecological distinctions between demographic and environmental stochasticity (Engen et al., 1998; Rohani et al., 2002).

Our inferential targets must be distinguished from the apparent reproduction numbers analysed by Eales and Riley (2024). Eales and Riley (2024) demonstrated that downstream surveillance time series (such as hospitalisations, intensive care admissions, or deaths) reconstruct an apparent reproduction number that is delayed and smoothed relative to the true transmission rate due to convolution with clinical delay distributions. In our framework, Target 1 is the conditional instantaneous transmission rate and Target 2 is the cross-period mean. Downstream surveillance outcomes need not reproduce the temporal dynamics of infection incidence without distortion (Eales and Riley, 2024). We therefore interpret our hospitalisation analysis as an empirical assessment of error scaling in a surveillance-derived growth statistic, rather than as a direct validation of the theoretical error decomposition for the latent reproduction number. While deconvolution techniques can mitigate apparent delay and smoothing, they cannot remove environmental variance from Target 2, because environmental fluctuations represent physical variation in transmission rather than an artefact of lagged observation.

The theoretical lower bounds established here apply strictly to single-period, conditionally unbiased estimators of R_t and to one-step forecasts

under independent environments. Estimators that tolerate conditional bias—such as James–Stein shrinkage, empirical Bayes methods (Efron and Morris, 1975), and state-space smoothers like EpiFilter (Parag, 2021)—can achieve a mean squared error below v_R by shrinking estimates toward a global mean $\bar{R}$. For example, under baseline parameters with $m = 50$, linear shrinkage achieves MSE below $v_R = 0.04$ whenever the prior mean is within 0.279 of $\bar{R}$, and achieves MSE of 0.0089 (22% of v_R) when the prior mean is within 0.1 of $\bar{R}$ (Supplementary Section S1.11). However, this variance reduction comes at the cost of conditional bias that peaks precisely when R_t deviates furthest from normal transmission, which is when outbreak detection is most urgent. Sizing surveillance under biased estimators requires explicit loss trade-offs that lie outside the conditionally unbiased framework.

For surveillance design, our three levers target distinct error components. Expanding single-period sample volume suppresses sampling variance, but its value diminishes rapidly once volume reaches the scale of $m_\times$. Accumulating observation periods or pooling across jurisdictions suppresses the environmental term; indeed, the estimation precision of v_R itself improves solely with period count T (Proposition 5). Standardising reporting protocols or recording institutional reporting completeness eliminates reporting heterogeneity v_H. In baseline calculations, quadrupling period count reduces total MSE by 72% (from 22.5×10^{-3} to 6.2×10^{-3}), whereas increasing single-period sample volume tenfold reduces MSE by 17% (to 18.7×10^{-3}). When environmental shocks exhibit positive autocorrelation, surveillance design must account for the budget threshold $\mathcal{B}^\dagger$ (Proposition 7): small-budget systems should prioritize fewer, larger observation windows to avoid severe under-sampling per generation.

Surveillance practice should decouple early warning from trend estimation. Outbreak detection ($R_t > 1$) is an instance of Target 1, where sampling volume maintains continuous value. At small scales, exact tests should replace Wald tests to prevent size distortion, and surveillance capacity must be sized according to detection power curves (Section 3.1) and deployed to sentinel facilities with rapid, complete reporting. Conversely, tracking baseline transmission trends represents Target 2; once capacity approaches $m_\times$, additional resources are far more effectively allocated to temporal replication, broader geographic pooling, or reporting standardisation.

Several methodological limitations warrant emphasis. First, weekly hospital admissions aggregate approximately two influenza generations, meaning empirical regression coefficients a and b capture composite scaling shapes

rather than structural single-generation biological parameters. Second, while model comparisons decisively favor models with non-vanishing error over floorless specifications, current sample sizes cannot fully separate a strictly flat asymptotic plateau from slow power-law decay ($\hat{p} = 0.273$). Third, empirical intercepts reflect unmodelled error components not explained by sample volume and cannot be uniquely attributed to environmental variance; our hospital reporting completeness proxy accounts for $< 1\%$ of intercept excess, but cannot capture within-hospital underreporting or behavioral shifts in care-seeking. Fourth, seasonal transmission variance is fundamentally constrained by the number of observed epidemic seasons: by Corollary 2, three seasons bound the relative standard error of seasonal variance from below by 0.82, precluding precise estimation of multi-year baseline stability without extended longitudinal archives.

5. Conclusion

The marginal value of surveillance case volume is governed by the interplay between the inferential target and intrinsic error floors. When monitoring current transmission potential, returns to sample volume do not saturate, and surveillance capacity should be determined by statistical detection power. When estimating cross-period transmission trends, expanding single-period case counts yields diminishing returns; once sample volume reaches the variance balance scale $m_\times$, additional resources should be redirected toward accumulating observation periods, expanding spatial pooling, standardizing reporting protocols, and regularly updating baseline benchmark models. Empirical analysis of US influenza hospitalisations confirms an error scaling pattern compatible with this theoretical decomposition, in which unmodelled error components account for approximately half of early-epidemic loss and cannot be explained by reporting completeness proxies, though current observations cannot separate a strict non-zero asymptote from slow decay. Resolving the scale of environmental variance requires accumulating observation periods across multiple epidemic seasons. Surveillance systems must therefore move beyond single-period sample volume as the sole metric of monitoring adequacy.

6. Materials and methods

6.1. Branching and observation models

The epidemiological branching and surveillance observation process is formulated in Section 2.1 under foundational Assumptions H1–H4 (binomial observation, homogeneous branching, exogenous overdispersion, and predetermined ancestor scale) and extended in Section 2.3 under Assumptions H5–H6 (stochastic reporting and conditional independence).

Assumption H4 holds naturally when n is fixed by active contact-tracing design. In passive surveillance, n_t is the dynamic realization of prior transmission; renewal process and dynamic autoregressive simulations (Sections 6.3 and 6.4) relax this independence. Assumption H5 accommodates both binomial sampling fluctuations and institutional reporting shifts, differing in whether σ^2 scales with ρ.

6.2. Theoretical derivations

For Target 2, an estimator $\tilde{\theta}$ is conditionally unbiased for R_t if $\mathbb{E}(\tilde{\theta} \mid R_t) = R_t$. Decomposing estimation error into $\tilde{\theta} - \bar{R} = (\tilde{\theta} - R_t) + (R_t - \bar{R})$, the cross-product vanishes in expectation under the law of total expectation:

$$\mathbb{E}(\tilde{\theta} - \bar{R})^2 = \mathbb{E}\big[\mathrm{Var}(\tilde{\theta} \mid R_t)\big] + v_R \geq v_R + \mathbb{E}\big[1/\mathcal{I}(R_t)\big] = v_R + \frac{A}{m}, \qquad (22)$$

where $\mathbb{E}[1/\mathcal{I}(R_t)] = \mathbb{E}[R_t/(\rho n) + R_t^2/(kn)] = A/m$ using $\mathbb{E}(R_t^2) = \bar{R}^2 + v_R$. The estimator $\widehat{R}_t = C_t/m$ achieves equality, establishing $v_R + A/m$ as the minimum mean squared error in this class.

Under H4, any predictor $\hat{y}$ forecasting $y = C_{t+1}/m$ from history $\mathcal{F}_t$ satisfies $\mathbb{E}(\hat{y} - y)^2 \geq \mathrm{Var}(y \mid \mathcal{F}_t) = v_R + A/m$, with equality achieved at $\hat{y} = \bar{R}$. When $\bar{R}$ is estimated from T_0 prior independent periods of equal scale m, forecast error scales as $(v_R + A/m)(1 + 1/T_0)$. Under stationary AR(1) environments with persistence ϕ, $\mathrm{Var}(R_{t+1} \mid R_t) = (1 - \phi^2)v_R$. Multi-period allocations and shrinkage derivations are detailed in Supplementary Sections S1.1–S1.11.

6.3. Renewal processes and growth statistics

In passive time-series surveillance, active ancestor volume n is unobserved. Secondary infections arise from the convolution of past incidence with the generation interval distribution w_s: $\Lambda_{t+1} = \sum_{s \geq 1} w_s I_{t+1-s}$ (Fraser,

2007; Cori et al., 2013). Because true incidence is partially observed, infection pressure is unobserved. Surveillance constructs the empirical proxy $D_t = \sum_{s\geq 1} w_s C_{t+1-s}$, yielding the surveillance growth statistic $G_t = C_{t+1}/D_t$. Setting $m = \rho\mathbb{E}(\Lambda_{t+1} \mid \mathcal{F}_t)$ and applying the law of total variance twice (Supplementary Section S1.12):

$$\frac{\mathrm{Var}(C_{t+1} \mid \mathcal{F}_t)}{[\rho\mathbb{E}(\Lambda_{t+1} \mid \mathcal{F}_t)]^2} = v_R + \frac{A}{m} + \mathrm{CV}^2(\Lambda_{t+1} \mid \mathcal{F}_t)\,(\bar{R}^2 + v_R). \tag{23}$$

When infection pressure is known, $\mathrm{CV}(\Lambda_{t+1} \mid \mathcal{F}_t) = 0$, recovering Eq. (9). In dynamic surveillance, infection pressure uncertainty introduces an additional scale-invariant variance component.

The empirical relative squared error loss is $L_t = [(G_t - \bar{R})/\bar{R}]^2$, which equals the relative forecast error $[(C_{t+1} - \bar{R}D_t)/(\bar{R}D_t)]^2$ (Bosse et al., 2023). Replacing m with D_t, expected loss follows the affine scaling model:

$$\mathbb{E}(L_t \mid D_t) \approx a + \frac{b}{D_t}, \qquad a = \frac{v_R}{\bar{R}^2}, \quad b = \frac{A}{\bar{R}^2}, \quad \frac{b}{a} = m_\times. \tag{24}$$

We refer to b/a as the empirical balance ratio.

Accounting for binomial sampling noise in the denominator $D_t = m(1+\delta)$, where $\mathbb{E}(\delta^2) = (1-\rho)\sum_s w_s^2/m$, second-order delta expansion yields the modified slope approximation:

$$b \approx \frac{A + (1-\rho)(\bar{R}^2 + v_R)\sum_s w_s^2}{\bar{R}^2}. \tag{25}$$

Equation (25) generates theoretical slope ratios $r_j = b_j/b_{\text{primary}}$ across alternative weighting vectors w, while intercept ratios $q_j = a_j/a_{\text{primary}}$ differentiate reporting mechanisms (Supplementary Section S1.12 and Table S2).

6.4. Simulation design

Monte Carlo verification of Propositions 1 and 2 evaluated 10 scale points $m \in [1, 1000]$ with 20,000 replications each (seed 20261013; $\bar{R} = 1.15, v_R = 0.04, k = 0.35, \rho = 0.25$). Detection power curves were calculated analytically via Eq. (18).

Renewal simulations employed a 4-week discrete generation kernel $w = (0.45, 0.35, 0.15, 0.05)$. Weekly R_t was drawn independently from a Gamma distribution with mean 1.15 and variance 0.04, with $I_t \sim \mathrm{NB}(k\Lambda_t, k/(k+R_t))$, $k = 0.35$, and $C_t \sim \mathrm{Bin}(I_t, 0.25)$. Initial incidence for the first 4 weeks was

set to m_0/ρ, evaluating 8 scale tiers $m_0 \in [6, 800]$ with 400 trajectories of 30 weeks each (seed 20261014; 26 evaluation points per trajectory).

Autoregressive simulations evaluated whether temporal autocorrelation in R_t accounts for empirical intercept magnitude. We set $R_t = \max(0.05, \mu_R + e_t)$, where e_t is a stationary Gaussian AR(1) process with variance v_R and persistence $\phi \in \{0, 0.3, 0.5, 0.7, 0.9, 1\}$, with weekly transmission and binomial reporting $\text{Bin}(I_t, 0.021)$ across 153 state-season configurations matching NHSN empirical early-phase starting scales (seed 20261022; 400 replications per cell).

6.5. Data sources and preparation

Primary data comprise weekly laboratory-confirmed influenza hospital admissions and facility reporting completeness percentages across 50 US states and Washington, D.C., from NHSN (2020–2026; N = 16,218 state-weeks) (Centers for Disease Control and Prevention, 2026). Missing counts (504 state-weeks) were assigned zero. Surveillance scale D_t was measured using a three-week lookback:

$$D_t = 0.65\, C_t + 0.25\, C_{t-1} + 0.10\, C_{t-2}, \tag{26}$$

with average lookback lag of 1.45 weeks (10.2 days). Restricting to $D_t \geq 5$ yielded 10,303 valid observations. Sensitivity analyses evaluated single-week weighting ($w^{(1)} = (1, 0, 0)$; $\sum w^2 = 1$; mean lag 1.0 week) and three-week equal weighting ($w^{(3)} = (1/3, 1/3, 1/3)$; $\sum w^2 = 0.333$; mean lag 2.0 weeks), compared to primary weighting ($\sum w^2 = 0.495$).

Three reporting regimes occurred during the study period: mandatory reporting from October 2020 to April 30, 2024 (Regime 1; N = 6,009; mean reporting completeness 91.1%); voluntary reporting from May 1 to October 31, 2024 (Regime 2; N = 420; mean 54.6%, median 45.7%); and resumed mandatory reporting from November 1, 2024 onward (Regime 3; N = 3,874; mean 91.8%).

The early expansion window spans three seasons: October 8 to November 26, 2022; October 14 to December 16, 2023; and November 9, 2024 to January 18, 2025 (N = 1,267 state-weeks). The peak window encompasses the national peak week in each season (December 3, 2022; December 30, 2023; February 8, 2025) plus seven subsequent weeks (N = 1,220).

6.6. Statistical inference and model comparison

Model (24) was fitted via feasible generalised least squares (FGLS) with weights $1/\hat{\mu}^2$, where $\hat{\mu}$ denotes fitted values from the preceding iteration, iterated to numerical convergence (Wooldridge, 2010). Uncertainty was quantified through 2,000 state-clustered bootstrap replications (resampling 51 states with replacement; seed 20261017) (Cameron et al., 2008; Cameron and Miller, 2015). Primary reporting presents 95% percentile and BCa confidence intervals (Efron, 1987), with acceleration estimated via leave-one-state-out jackknife.

Model comparisons on mean functional shape evaluated three pre-specified specifications: $L = b/D$ (floorless), $L = a + b/D$ (affine), and $L = cD^{-p}$ (power law). Models were estimated by Gamma quasi-likelihood with variance proportional to the square of the mean (McCullagh and Nelder, 1989). Selection criteria included AIC and leave-one-state-out cross-validation deviance, re-estimating $\bar{R}$ solely on training states in each fold.

Two-stage instrumental variable regressions instrumented $1/D_t$ with lagged scales $D_{t-3}, \ldots, D_{t-5}$, evaluated with 10,000 bootstrap resamples. Historical benchmark models compared expanding-window, 26-week rolling, and state-specific expanding estimates using only antecedent data.

6.7. Reporting adjustment and cross-pathogen analysis

Hospital reporting completeness is denoted by $\hat{p}_{s,t}$. By Proposition 3, time-invariant reporting cancels from the growth ratio; the variance entering the intercept is $V_\pi = \mathrm{Var}(\pi_{s,t})$, where $\pi_{s,t} = \hat{p}_{s,t+1} / \sum_j w_j \hat{p}_{s,t-j}$. Reporting-adjusted counts are $C^*_{s,t} = C_{s,t}/\hat{p}_{s,t}$, yielding adjusted denominator D^*_t and growth ratio $G^*_t = G_t/\pi_{s,t}$. FGLS regression of L^*_t on $1/D_t$ yields adjusted intercept a^* and difference $\Delta a = a - a^*$. Two one-sided tests for equivalence (TOST) evaluated equivalence within a $\pm 5\%$ margin (± 0.0050) using 90% BCa confidence intervals.

Cross-pathogen replication evaluated COVID-19 and RSV admissions from NHSN across calendar windows and growth-phase windows ($G^{\mathrm{nat}}_t \geq 1.05$). Benchmark excess was computed as $a - 0.08/\bar{R}^2$. Rejection regions against dynamic growth nulls were established via 108 simulation cells of daily renewal processes ($\phi = 0$; varying overdispersion, generation interval, admission delay, and growth rate; seed 20261058) and 48 cells of multiplicative reporting shocks (seed 20261059). An outcome was graded as full replication if both q_1, q_2 excluded the null range and excess intercept lower bound

exceeded zero; partial replication if either criterion was met; and directionally consistent if intercept ratios excluded unity in the $q_1 < 1 < q_2$ direction.

CRediT authorship contribution statement

Chi Wang: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing - original draft, Writing - review & editing.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

All surveillance data used in this study are publicly available from the US Centers for Disease Control and Prevention National Healthcare Safety Network (NHSN), FluSight, ILINet, FluSurv-NET, and HHS Protect. Complete analytical scripts, simulation code, and processed panel datasets required to replicate all figures and numerical findings are archived in the study GitHub repository (https://github.com/wangchi2068/case-count-bottleneck).

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